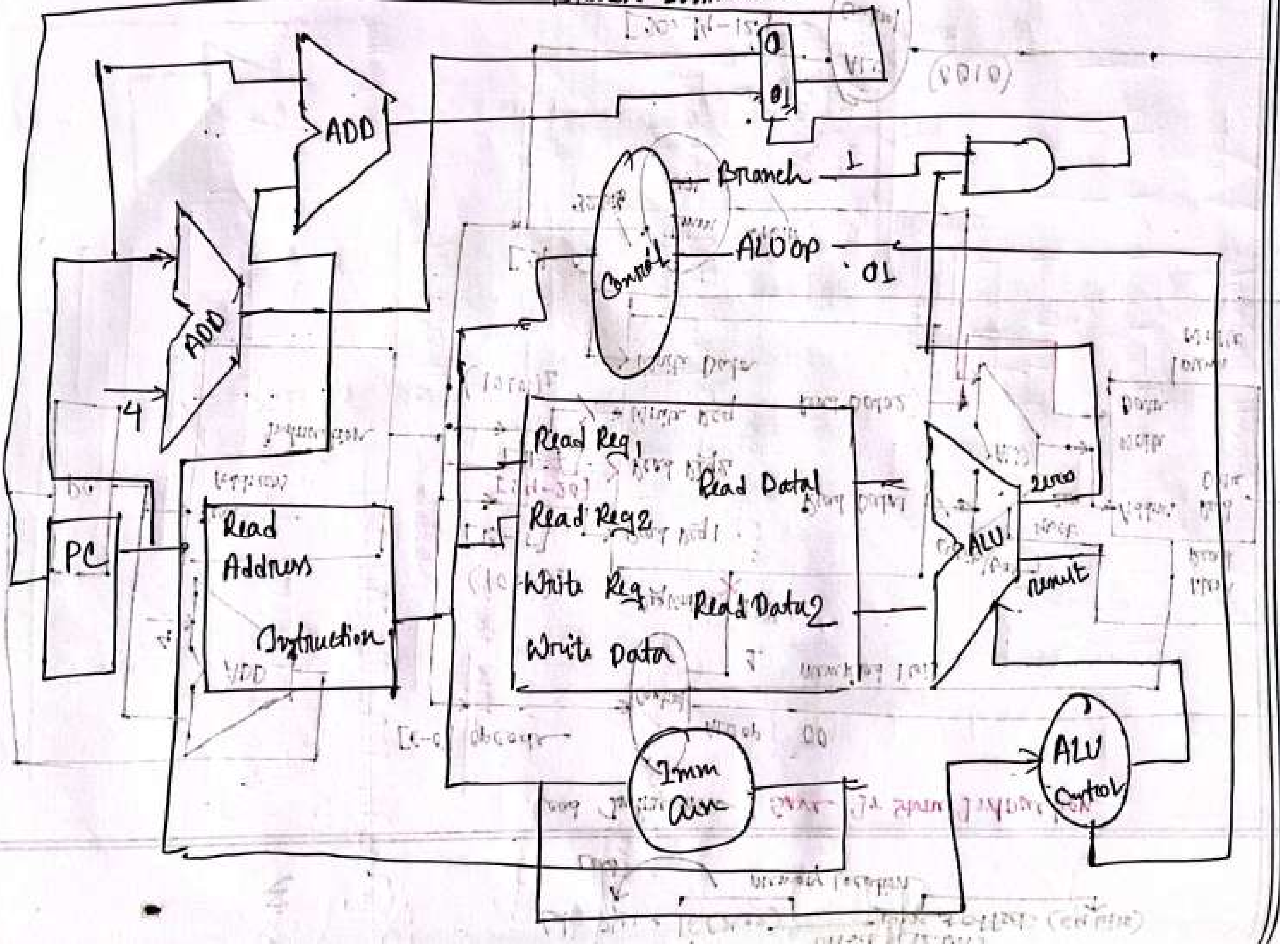
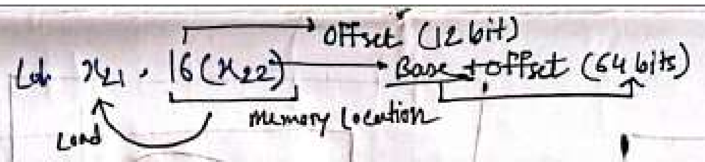
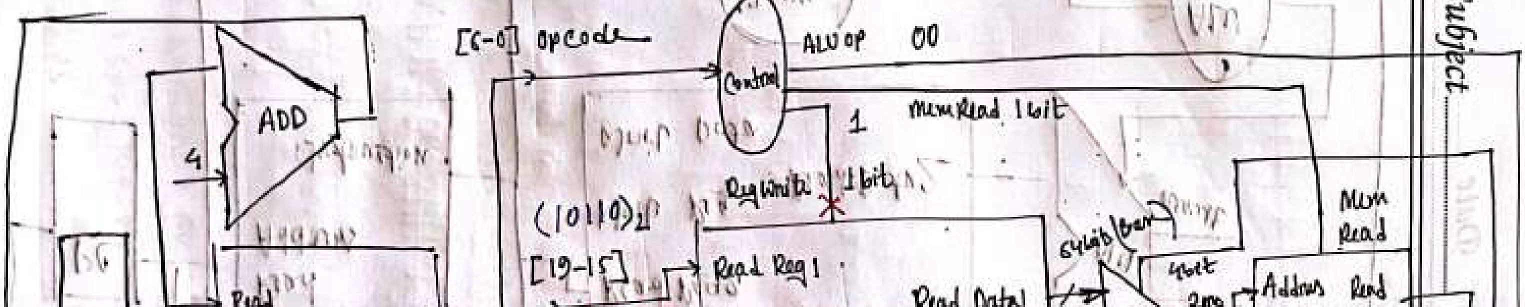
Branch Instruction


ADD
ADD
4
PC
Read Address
Instruction
0
01
Control
Branch
1
ALUOp
01
Read Reg1
Read Reg2
Write Reg
Write Data
Read Data1
Read Data2
ALU
zero
result
Imm Gen
ALU Control

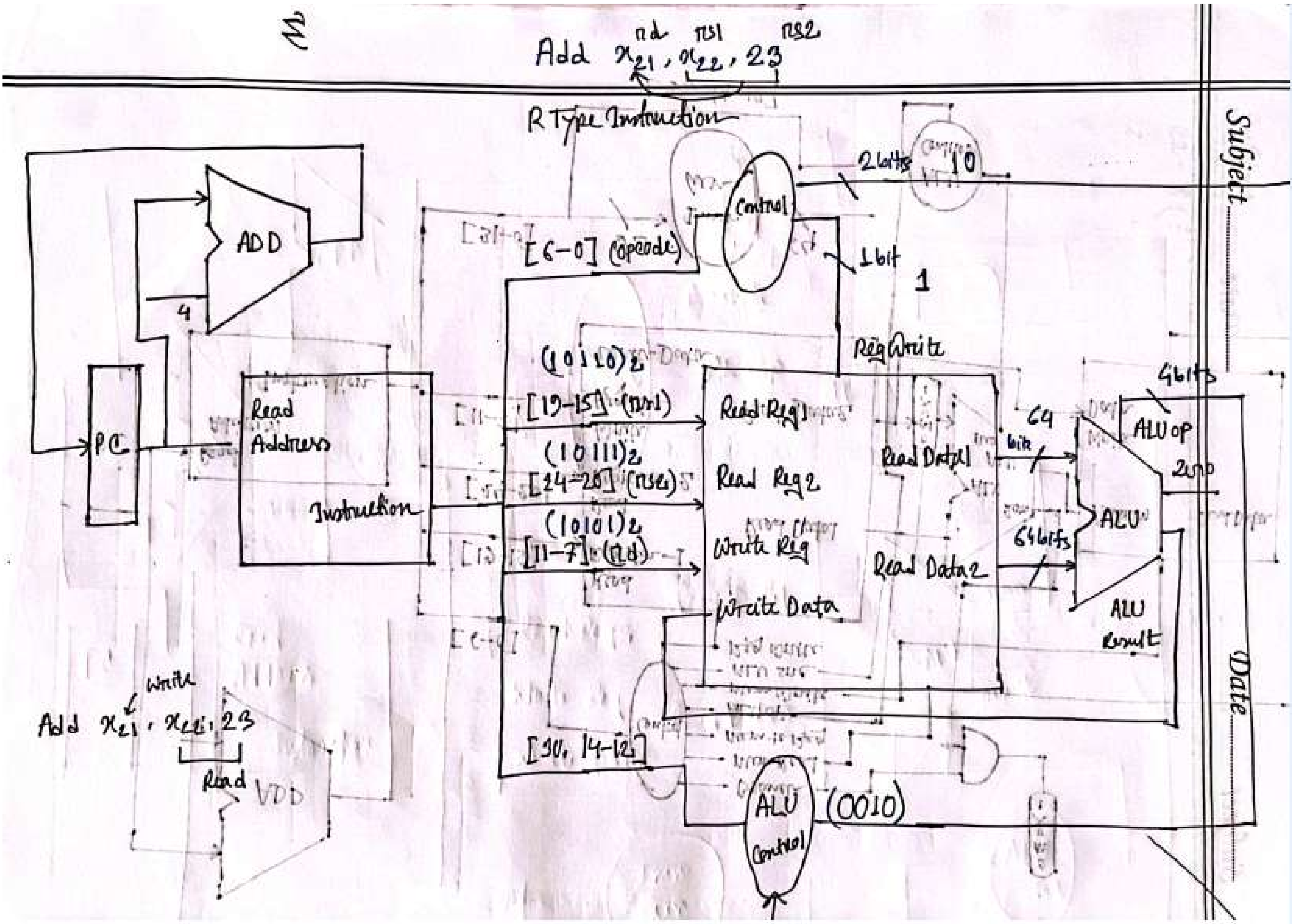

ld x21, 16(x22)
Offset (12 bit)
Base + offset (64 bits)
memory location
Load

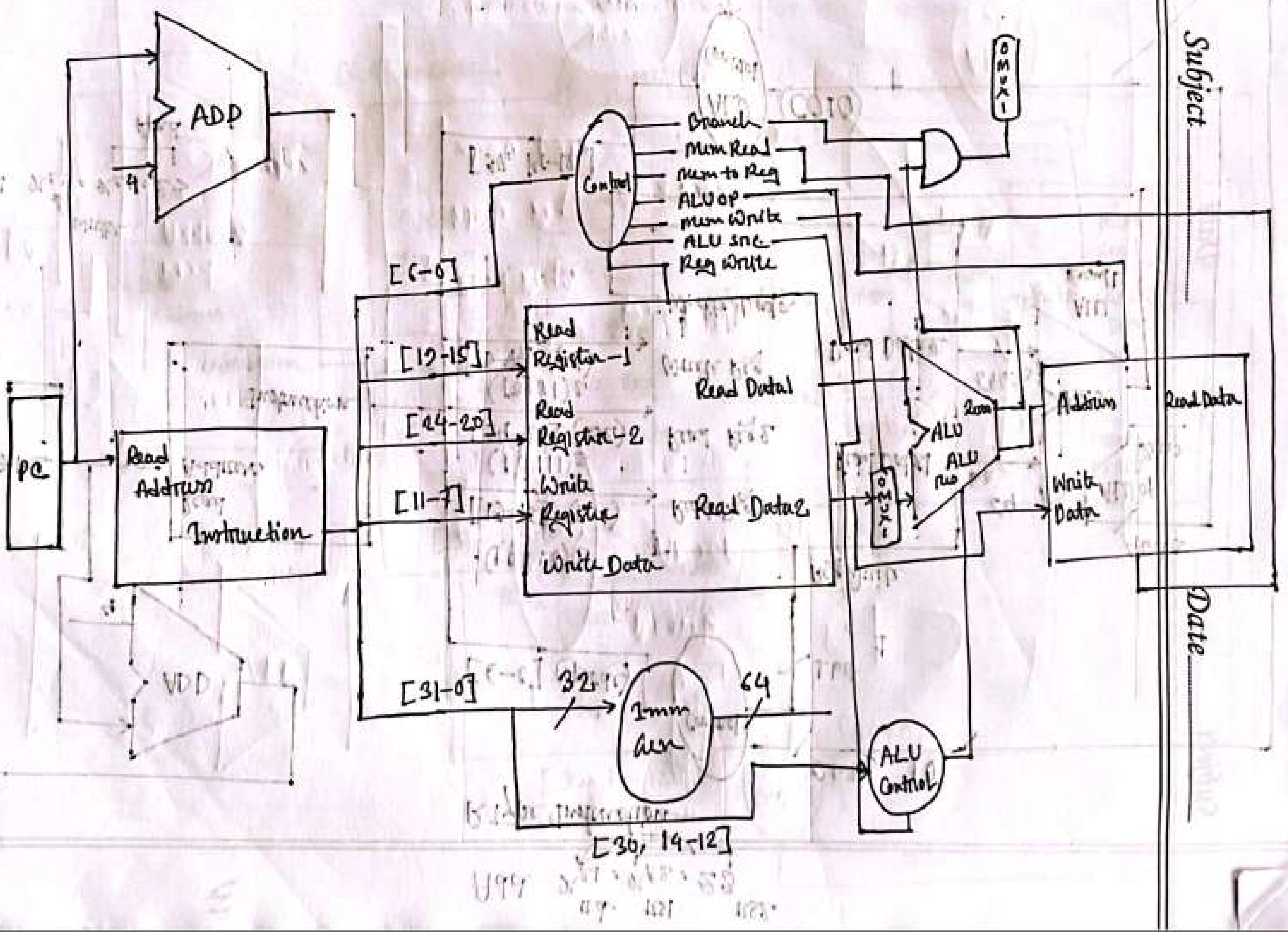
Load Instruction / ~~Save~~ Or Store Instruction


[6-0] opcode
Control
ALU op 00
memRead 1bit
1
RegWrite 1bit
ADD
4
PC
Read Address
Instruction
(10110)2
[19-15]
[24-20]
[11-7]
(10101)2
Read Reg1
Read Reg2
Write Reg
Write Data
Read Data1
Read Data2
64bits (base)
4bit
zero
ALU
Mem Read
Address
Read Data
Write Data
Mem Write
[31-0]
32bits
Imm Gen
64bits
ALU Control
(0010)
[30, 14-12]
(00)

Subject

Date


rd
rs1
rs2
Add x21, x22, 23
R Type Instruction
ADD
4
PC
Read Address
Instruction
[6-0] (opcode)
Control
2 bits
10
1 bit
1
RegWrite
(10110)2
[19-15] (rs1)
(10111)2
[24-20] (rs2)
(10101)2
[11-7] (rd)
Read Reg1
Read Reg2
Write Reg
Write Data
Read Data1
Read Data2
64 bit
64 bits
4 bits
ALU op
2 bit
ALU
ALU Result
[30, 14-12]
ALU Control
(0010)
Write
Add x21, x22, 23
Read

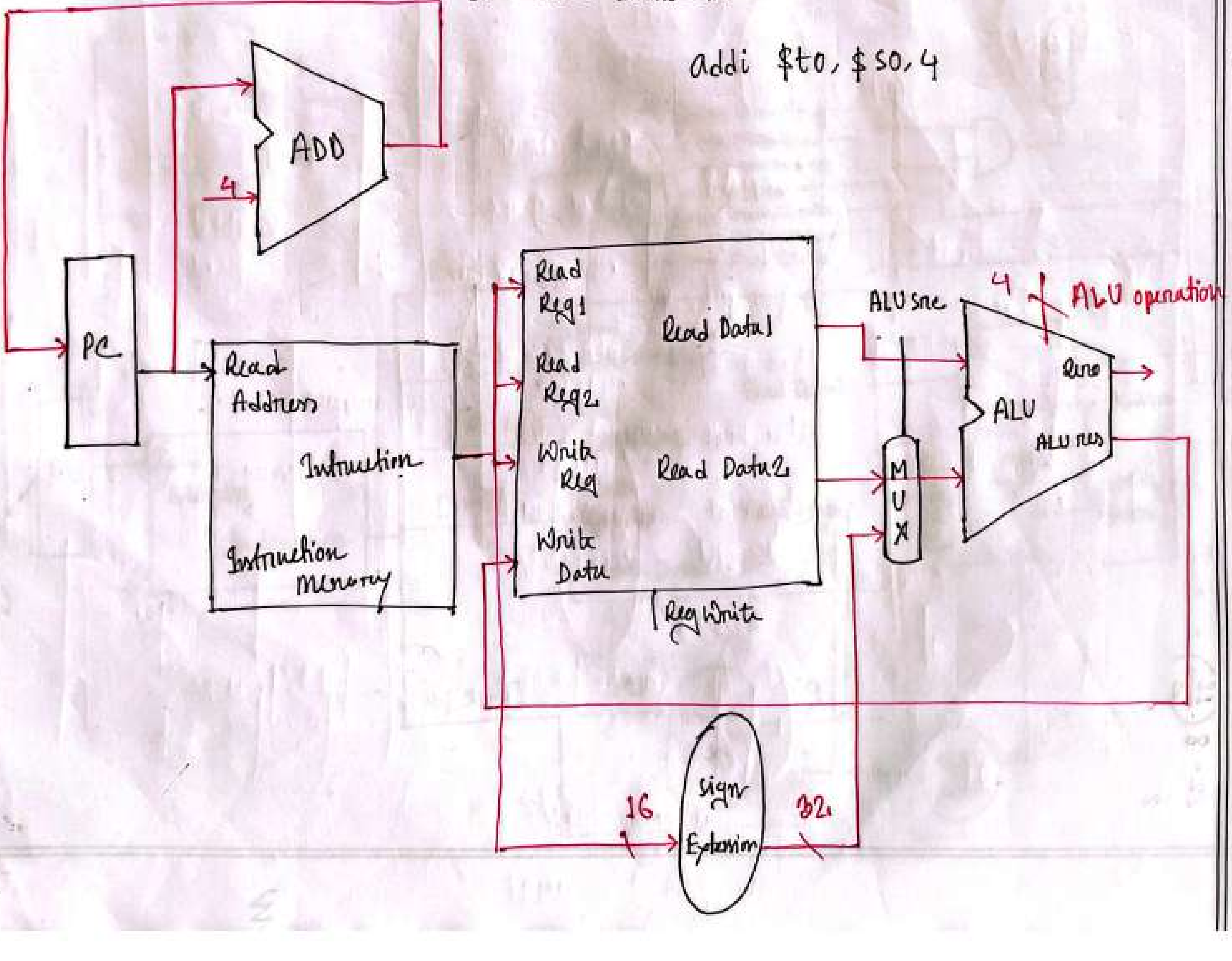

ADD
4
PC
Read Address
Instruction
Control
Branch
Mem Read
Mem to Reg
ALU op
Mem Write
ALU src
Reg Write
0 M U X 1
[6-0]
[19-15]
[24-20]
[11-7]
Read Register-1
Read Register-2
Write Register
Write Data
Read Data1
Read Data2
0 M U X 1
ALU
zero
ALU res
Address
Read Data
Write Data
[31-0]
32
Imm Gen
64
ALU Control
[30, 14-12]

Subject

Date

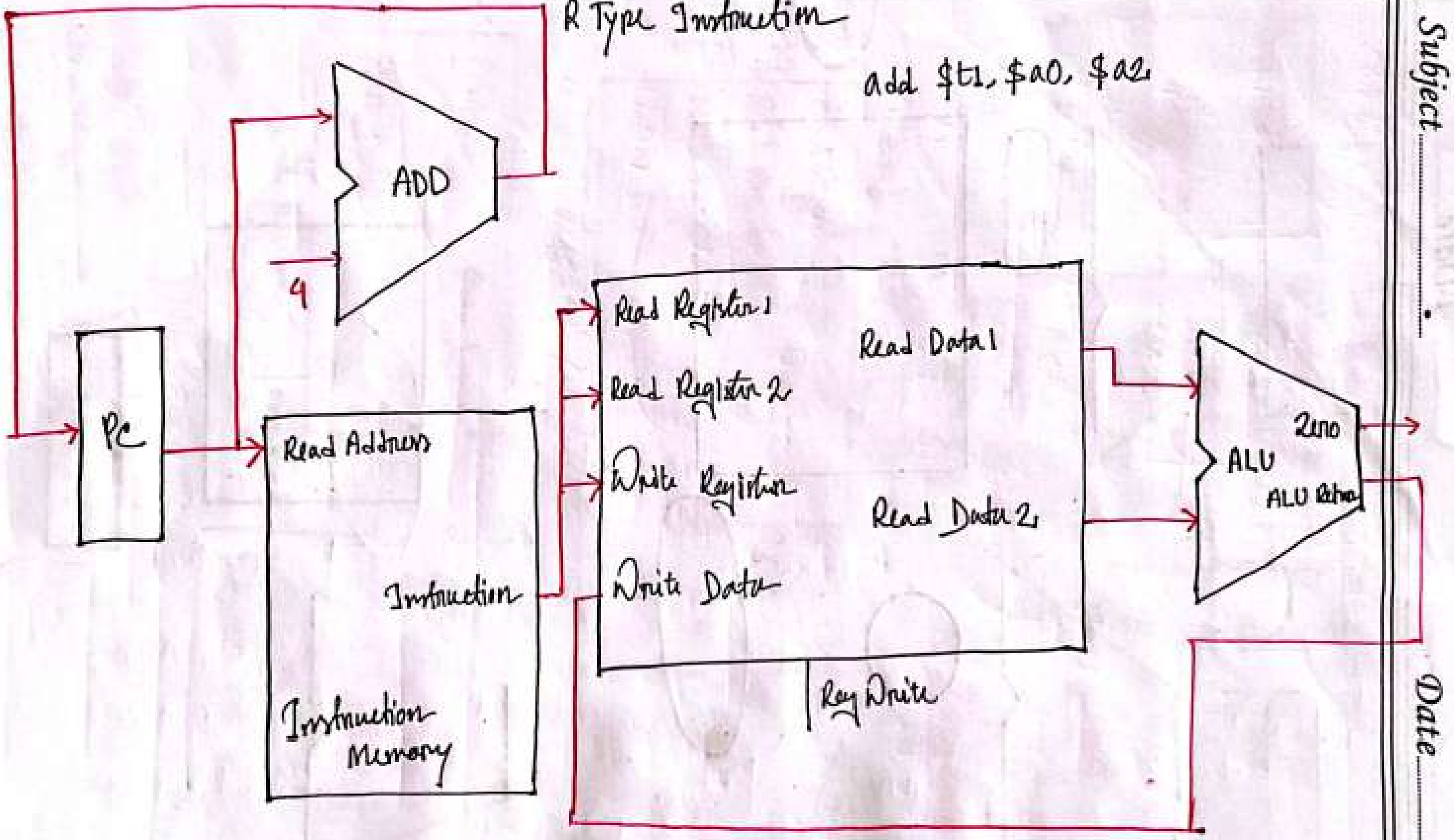
Immediate Instruction

addi $t0, $s0, 4

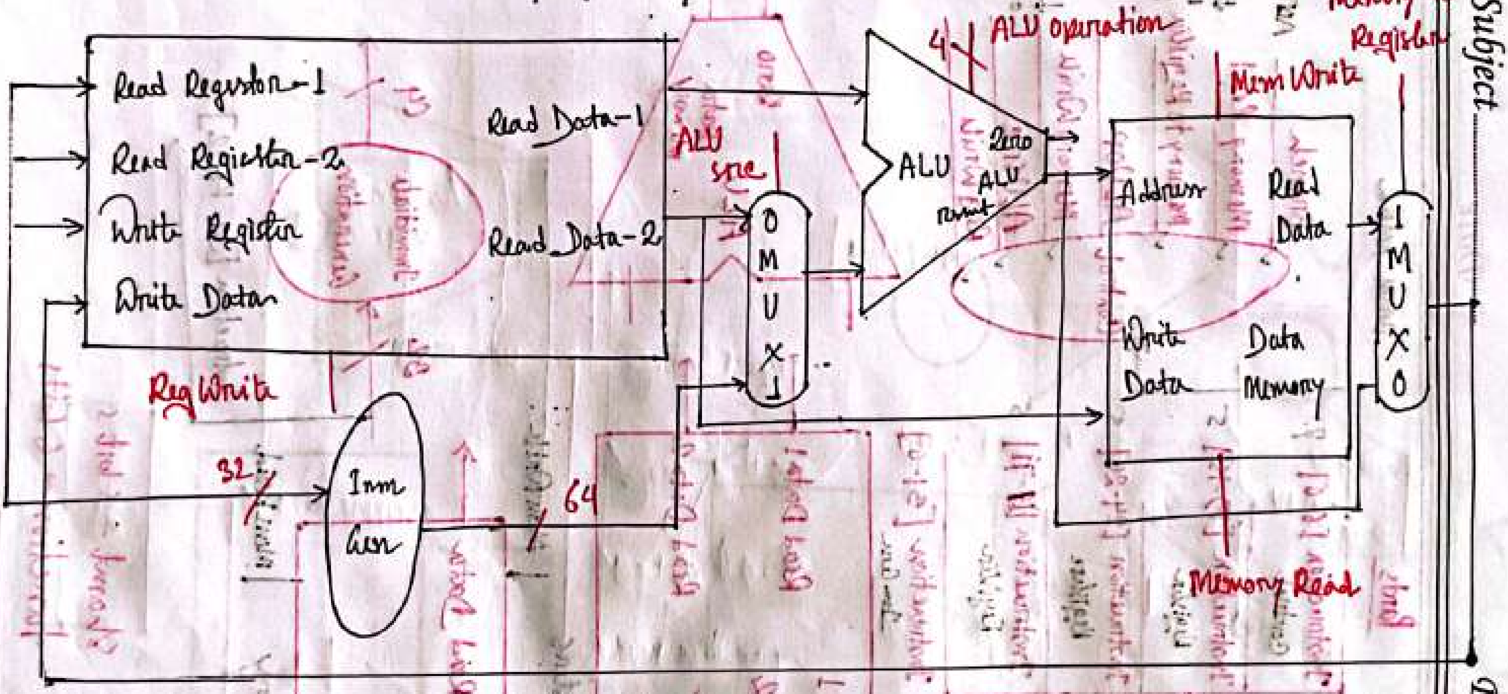

ADD
4
PC
Read Address
Instruction
Instruction memory
Read Reg1
Read Reg2
Write Reg
Write Data
Read Data1
Read Data2
RegWrite
ALUSrc
MUX
4
ALU operation
ALU
zero
ALU res
16
Sign Extension
32

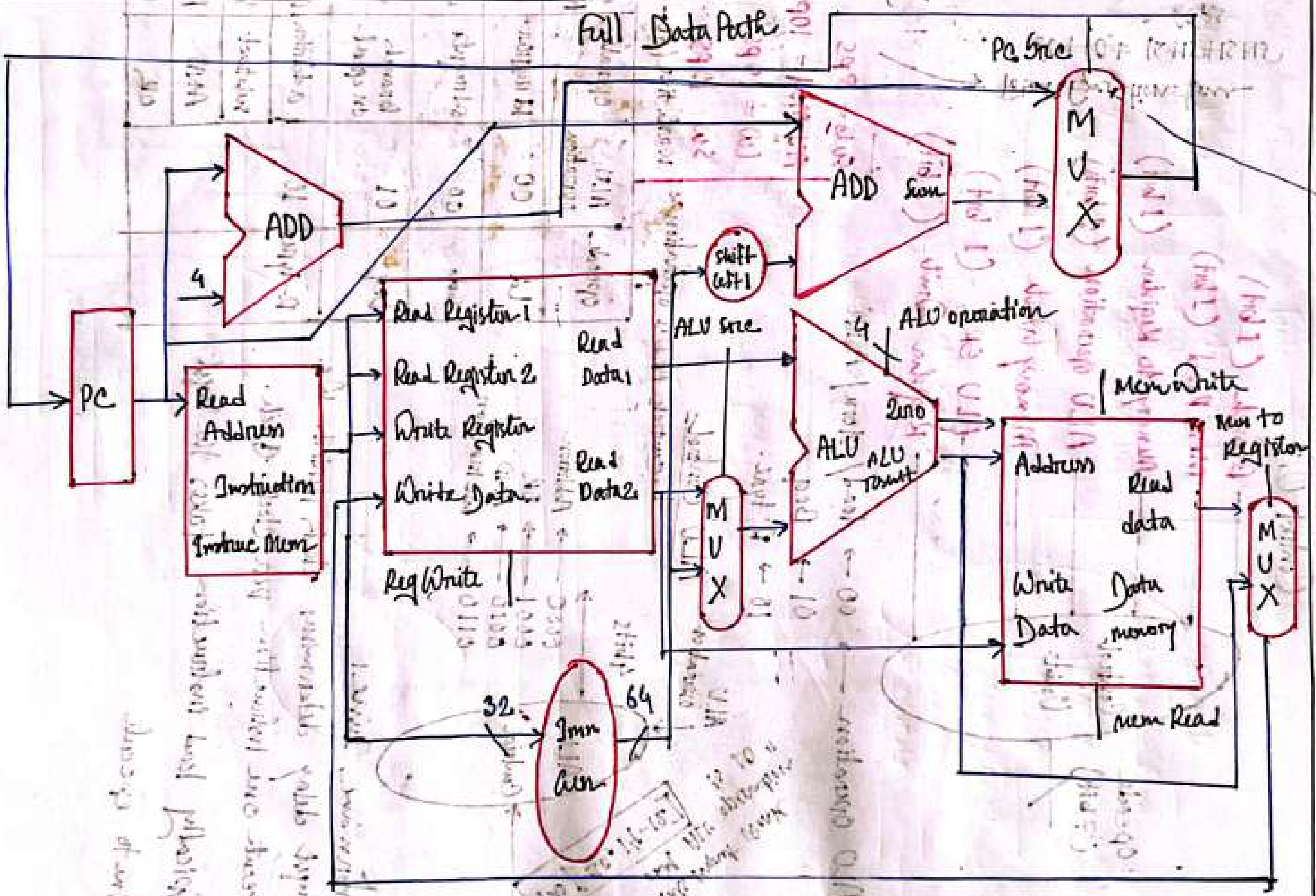
R Type Instruction

add $t1, $a0, $a2


ADD
4
PC
Read Address
Instruction
Instruction Memory
Read Register 1
Read Register 2
Write Register
Write Data
Read Data 1
Read Data 2
Reg Write
ALU
Zero
ALU Result

R Type / Load / Store Datapath


Read Register-1
Read Register-2
Write Register
Write Data
Read Data-1
Read Data-2
RegWrite
32
Imm Gen
64
ALU src
0 MUX 1
4
ALU operation
ALU
Zero
ALU result
Mem Write
Address
Read Data
Write Data
Data Memory
Memory Read
Memory to Register
1 MUX 0


Full Data Path
PC
ADD
4
Read Address
Instruction
Instruc Mem
Read Register 1
Read Register 2
Write Register
Write Data
Read Data1
Read Data2
RegWrite
32
Imm Gen
64
Shift Left 1
ADD
Sum
PC Src
M U X
ALU Src
M U X
4
ALU operation
ALU
Zero
ALU result
Mem Write
Address
Read data
Write Data
Data memory
Mem Read
Mem to Register
M U X